

Practice Final Test-Part I (Version 1)

Name
Date
Course

- I) This exam covers the first half of the course (everything up to test II).
 II) Write your name, the date, and the course title (i.e. Precalculus).
 III) Show all work.
 IV) Draw a rectangle around your final answer for each problem.
 V) You may use a calculator for counting problems only, not for graphing.

① A polynomial function is written below in factored form. Write the x-intercepts of the corresponding graph and determine if the function crosses or touches the x-axis at each intercept.

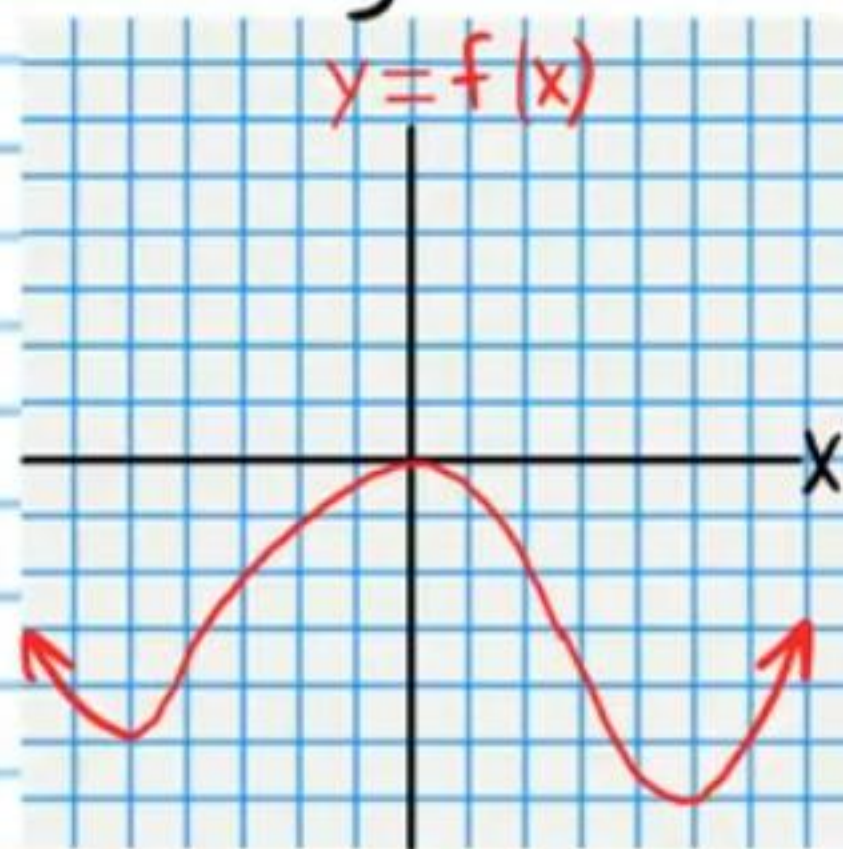
$$f(x) = -3(x-2)^4(x+5)^3(x-\pi)$$

2	⇒ Touches
-5	⇒ Crosses
π	⇒ Crosses

② The graph of a function is written below. Write the intervals where the function is increasing or decreasing.

Increasing: $(-5, 0) \cup (5, 7)$

Decreasing: $(-7, -5) \cup (0, 5)$



③ Lydia has a coin collection. The probability that a randomly chosen coin will be made of silver is $\frac{1}{3}$. The probability that a randomly chosen coin will be a dime is $\frac{4}{9}$. The probability that a randomly chosen coin will be both made of silver and a dime is $\frac{1}{5}$. What is the probability that a randomly chosen coin will be either made of silver or a dime?

$$\begin{aligned} P(S \cup d) &= P(S) + P(d) - P(S \cap d) \\ &= \frac{1}{3} + \frac{4}{9} - \frac{1}{5} \\ &= \frac{3}{9} + \frac{4}{9} - \frac{1}{5} \\ &= \frac{7}{9} - \frac{1}{5} \\ &= \boxed{\frac{26}{45}} \end{aligned}$$

④ What is the probability that a randomly selected coin from Lydia's collection will not be made of silver?

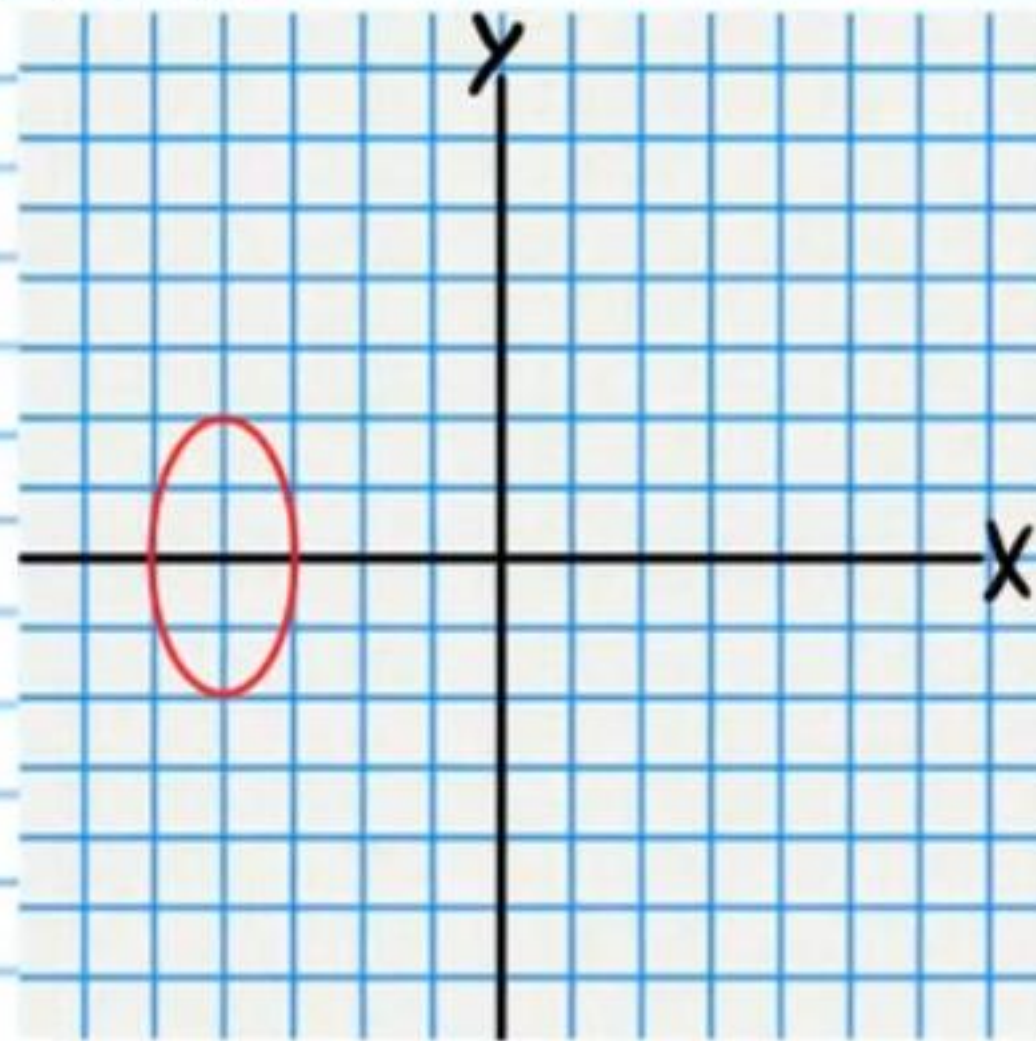
$$\begin{aligned} P(\text{Not silver}) &= 1 - P(\text{silver}) \\ &= 1 - \frac{1}{3} \\ &= \boxed{\frac{2}{3}} \end{aligned}$$

⑤ Write the equation of the graph below.

$$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$$

$$\frac{(x-(-4))^2}{1^2} + \frac{(y-0)^2}{2^2} = 1$$

$$\boxed{(x+4)^2 + \frac{y^2}{4} = 1}$$



⑥ There are 28 letters in the arabic alphabet. How many ways can 6 letters be permuted (i.e. ordered) if repetition is allowed (i.e. letters can be chosen more than once)?

$$28 \times 28 \times 28 \times 28 \times 28 \times 28 = 28^6$$

$$= \boxed{481,890,304 \text{ outcome variations}}$$

⑦ Alexander will give one card to each of his four children. He is using a deck with 50 distinct cards. How many outcome variations are possible?

Note: A scenario where Cecelia gets the ace of diamonds is different from a scenario where she gets the 8 of clubs.

$$P(50, 4) = {}_{50}P_4$$

$$= 50 \times 49 \times 48 \times 47 = \boxed{5,527,200 \text{ outcome variations}}$$

⑧ Find an equation of the circle with center on the line $x=2$, radius 2, and point on the circle $(4, 3)$. Then graph the equation.

$$(x-h)^2 + (y-k)^2 = r^2$$

$$(4-2)^2 + (3-k)^2 = 2^2$$

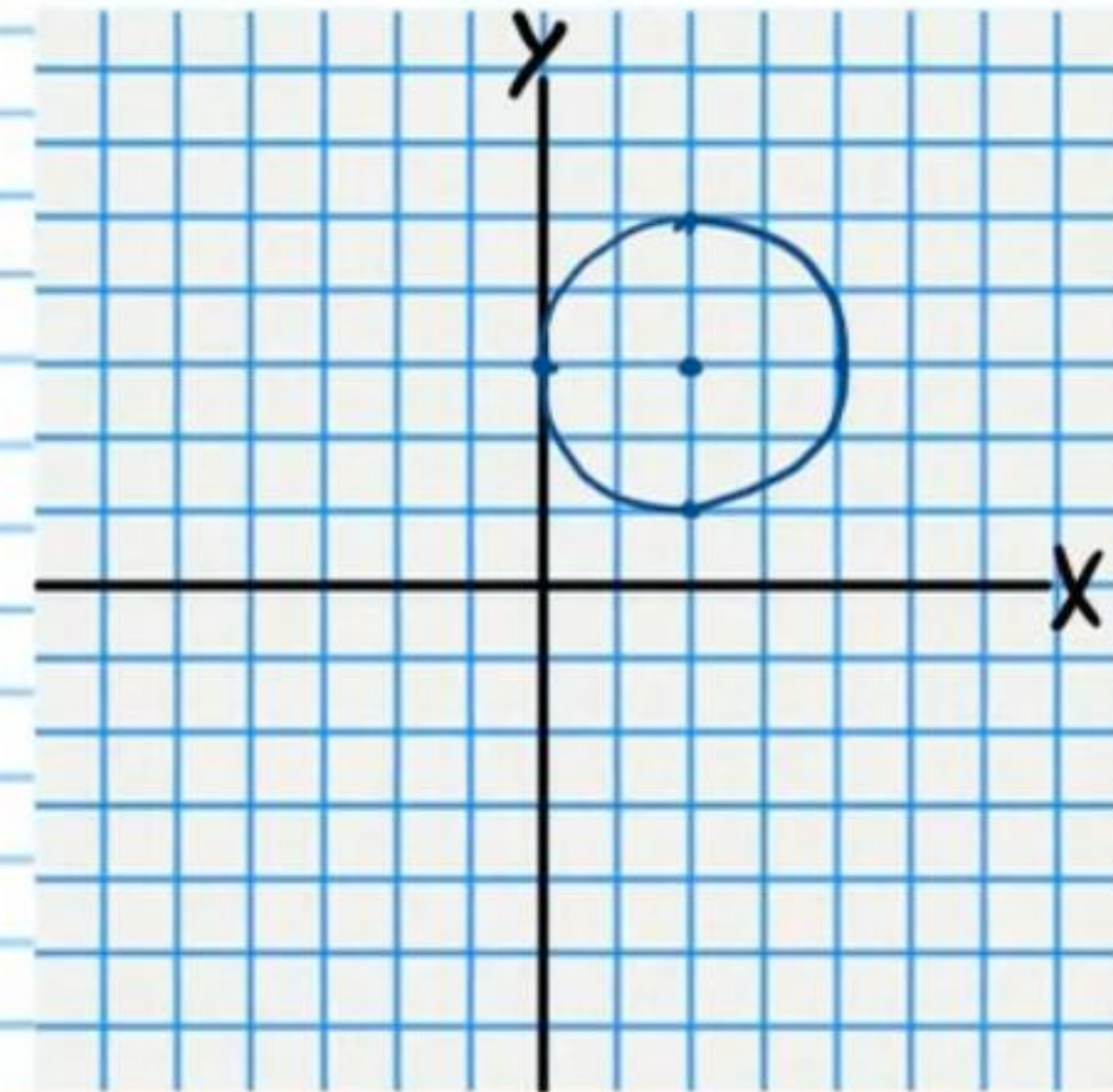
$$2^2 + (3-k)^2 = 2^2$$

$$(3-k)^2 = 0$$

$$3-k = \pm\sqrt{0}$$

$$k = 3$$

$$\boxed{(x-2)^2 + (y-3)^2 = 4}$$



⑨ i) Write an equation of the n^{th} term of the sequence below in recursive form (show the value of the first term).

$$-\frac{1}{2}, \frac{1}{4}, -\frac{1}{8}, \dots$$

$$\boxed{\{a_n = -\frac{1}{2}a_{n-1}, a_1 = -\frac{1}{2}\}}$$

ii) What is the sum of all of the terms?

$$S_{\infty} = \frac{a_1}{1-r}$$

$$S_{\infty} = \frac{-\frac{1}{2}}{1-(-\frac{1}{2})} = \frac{-\frac{1}{2}}{1+\frac{1}{2}} = \frac{-\frac{1}{2}}{\frac{3}{2}} = -\frac{1}{2} \div \frac{3}{2}$$

$$= \boxed{-\frac{1}{3}}$$

iii) What is the 30th term?
You may leave your answer in exponential form.

$$a_n = a_1 r^{n-1}$$

$$a_{30} = -\frac{1}{2} \left(-\frac{1}{2}\right)^{30-1}$$

$$= -\frac{1}{2} \left(-\frac{1}{2}\right)^{29}$$

$$= \boxed{\left(-\frac{1}{2}\right)^{30}}$$

iv) What is the sum of the first 30 terms? You may leave your answer in exponential form.

$$S_n = a_1 \frac{1-r^n}{1-r}$$

$$S_{30} = \left(-\frac{1}{2}\right) \frac{1 - \left(-\frac{1}{2}\right)^{30}}{1 - \left(-\frac{1}{2}\right)}$$

$$S_{30} = \left(-\frac{1}{2}\right) \frac{1 - \left(\frac{1}{2}\right)^{30}}{\frac{3}{2}}$$

$$S_{30} = \boxed{\frac{-1 + \left(\frac{1}{2}\right)^{30}}{3}}$$

⑩ Use the Binomial Theorem to expand the following expression.

$$(x^3 - 2)^4$$

$$\binom{4}{0}(x^3)^4(-2)^0 + \binom{4}{1}(x^3)^3(-2)^1 + \binom{4}{2}(x^3)^2(-2)^2 + \binom{4}{3}(x^3)^1(-2)^3 + \binom{4}{4}(x^3)^0(-2)^4$$

$$(1)x^{12}(1) + 4(x^9)(-2) + (6)(x^6)(4) + (4)x^3(-8) + (1)x^0(16)$$

$$\boxed{x^{12} - 8x^9 + 24x^6 - 32x^3 + 16}$$

⑪ Find the value of the expression below.

$$\sum_{n=2}^{72} 3n-1 = (3(2)-1) + (3(3)-1) + (3(4)-1) + (3(5)-1) + \dots$$

$$= 5 + 8 + 11 + 14 + \dots$$

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

$$S_{71} = \frac{71}{2}(2(5) + 3(71-1))$$

$$= \frac{71}{2}(10 + 3(70))$$

$$= \frac{71}{2}(10 + 210)$$

$$= \frac{71}{2}(220)$$

$$= (71)(110)$$

$$= \boxed{7,810}$$

⑫ Write a polynomial function in standard form with real coefficients that has the following zeros, leading coefficient (LC), and degree (D).

Zeros: $i, 2$ $LC = -1$ $D = 3$

$$\begin{aligned}
 f(x) &= -(x-2)(x-i)(x+i) \\
 &= -(x-2)(x^2 + i x - i x - i^2) \\
 &= -(x-2)(x^2 - (-1)) \\
 &= -(x-2)(x^2 + 1) \\
 &= -(x^3 + x - 2x^2 - 2) \\
 &= \boxed{-x^3 + 2x^2 - x + 2}
 \end{aligned}$$

⑬ i) Divide.

$$(-x^2 + x^5 + 3) \div (2x - 1 + x^2)$$

$$\begin{array}{r}
 x^3 - 2x^2 + 5x - 13 \\
 x^2 + 2x - 1 \overline{) x^5 - x^2 + 3} \\
 \underline{x^5 + 2x^4 - x^3} \\
 -2x^4 + x^3 - x^2 + 3 \\
 \underline{-2x^4 - 4x^3 + 2x^2} \\
 5x^3 - 3x^2 + 3 \\
 \underline{5x^3 + 10x^2 - 5x} \\
 -13x^2 + 5x + 3 \\
 \underline{-13x^2 - 26x + 13} \\
 31x - 10
 \end{array}$$

$$x^3 - 2x^2 + 5x - 13 + \frac{31x - 10}{x^2 + 2x - 1}$$

ii) Can you use synthetic division? Explain.

No. The divisor is not a binomial of the form $x - a$.

⑭ Use the Rational Zeros Theorem to find the zeros (real or imaginary) of the polynomial function below and write the multiplicity of each zero.

$$f(x) = 2x^4 - 9x^3 + 14x^2 - 12x + 8$$

$$p: 8 \Rightarrow 1, 2, 4, 8$$

$$q: 2 \Rightarrow 1, 2$$

$$\frac{p}{q}: \pm 1, \pm 2, \pm 4, \pm 8, \pm \frac{1}{2}$$

$$1 \Rightarrow \text{X}$$

$$-1 \Rightarrow \text{X}$$

$$2 \Rightarrow \checkmark$$

$$2 \Rightarrow \checkmark$$

$$\begin{array}{r|rrrrrr} 2 & 2 & -9 & 14 & -12 & 8 \\ & & 4 & -10 & 8 & -8 \\ \hline & 2 & -5 & 4 & -4 & 0 \end{array}$$

$$2x^3 - 5x^2 + 4x - 4$$

$$\begin{array}{r|rrrr} 2 & 2 & -5 & 4 & -4 \\ & & 4 & -2 & 4 \\ \hline & 2 & -1 & 2 & 0 \end{array}$$

$$f(x) = (x-2)(x-2)(2x^2 - x + 2)$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$f(x) = (x-2)^2 \left(x - \frac{1+i\sqrt{15}}{4}\right) \left(x - \frac{1-i\sqrt{15}}{4}\right)$$

$$x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(2)(2)}}{2(2)}$$

$$x = \frac{1 \pm \sqrt{1-16}}{4}$$

$$x = \frac{1 \pm \sqrt{-15}}{4}$$

$$\begin{array}{l} 2 \Rightarrow M = 2 \\ \frac{1+i\sqrt{15}}{4} \Rightarrow M = 1 \\ \frac{1-i\sqrt{15}}{4} \Rightarrow M = 1 \end{array}$$

⑮ Indicate the end behavior of the function from problem #14.

$$\text{As } x \rightarrow \infty, f(x) \rightarrow \infty$$

$$\text{As } x \rightarrow -\infty, f(x) \rightarrow \infty$$

⑯ State the maximum number of turning points of the graph of the function in problem #14.

$$4 - 1 = \boxed{3}$$

⑰ There are 8 blue marbles, 5 green marbles, and 7 red marbles in a bag. 3 marbles are chosen at random.

i) What is the probability that all three are blue?

$$P(BBB) = \frac{{}^8C_3}{{}^{20}C_3} = \frac{56}{1,140} = \boxed{\frac{14}{285}}$$

ii) What is the probability that 2 marbles will be green and 1 will be blue?

$$\begin{aligned} P(2 \text{ green } 1 \text{ blue}) &= \frac{{}^5C_2 \times {}^8C_1}{{}^{20}C_3} = \frac{10 \times 8}{1,140} = \frac{80}{1,140} \\ &= \frac{8}{114} \\ &= \boxed{\frac{4}{57}} \end{aligned}$$

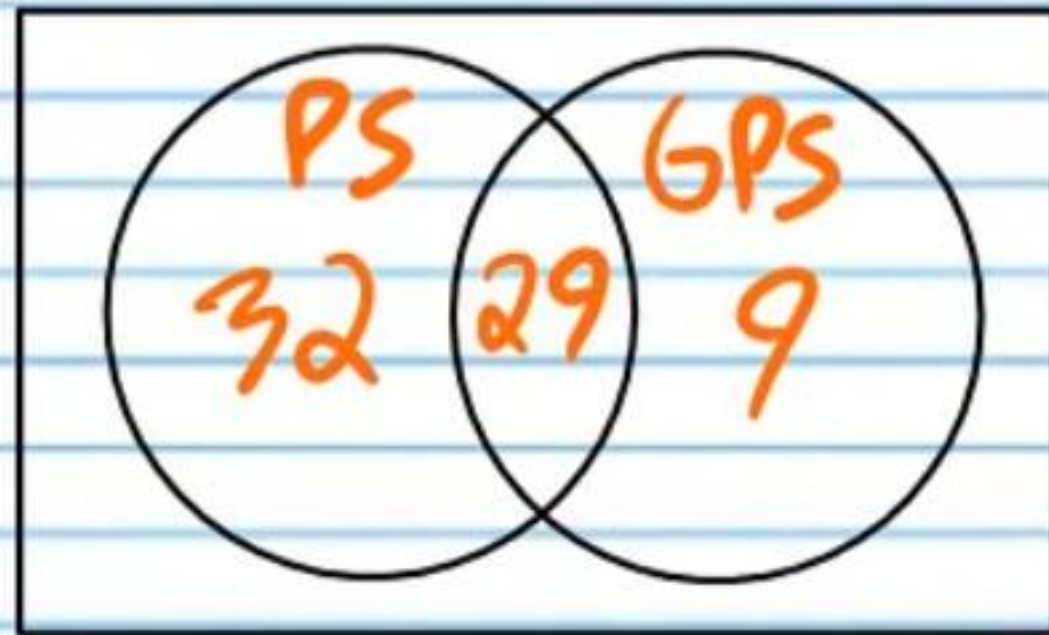
A factory produces 150 trucks in one month. 61 of the trucks have a proximity sensor. 38 of the trucks have a GPS tracker. 29 of the trucks have both a proximity sensor and a GPS tracker.

(18) i) How many trucks have a GPS tracker but not a proximity sensor?

$$38 - 29 = \boxed{9 \text{ trucks}}$$

ii) How many trucks have a proximity sensor but not a GPS tracker?

$$61 - 29 = \boxed{32 \text{ trucks}}$$



iii) How many trucks have **either** a proximity sensor **or** a GPS tracker?

$$32 + 29 + 9 = \boxed{70 \text{ trucks}}$$

iv) How many trucks have **neither** a proximity sensor **nor** a GPS tracker?

$$150 - 70 = \boxed{80 \text{ trucks}}$$

(19) i) What is the probability that a randomly selected truck will have a proximity sensor?

$$\frac{61}{150}$$

ii) What is the probability that a randomly selected truck will have **neither** a proximity sensor **nor** a GPS tracker?

$$\frac{80}{150} = \frac{8}{15}$$

iii) What is the probability that a randomly selected truck will have **both** a proximity sensor **and** a GPS tracker?

$$\frac{29}{150}$$

iv) What is the probability that a randomly selected truck will have **either** a proximity sensor **or** a GPS tracker?

$$\frac{70}{150} = \frac{7}{15}$$

Factoring the Sum or Difference of Two Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

20) Graph the equation below. Write the coordinates of the foci and the equations of the asymptotes.

$$4x^2 - 6y^2 - 8x + 24y - 44 = 0$$

$$4x^2 - 8x - 6y^2 + 24y = 44$$

$$4(x^2 - 2x) - 6(y^2 - 4y) = 44$$

$$4(x^2 - 2x + 1) - 6(y^2 - 4y + 4) = 44 + 4 - 24$$

$$4(x-1)^2 - 6(y-2)^2 = 24$$

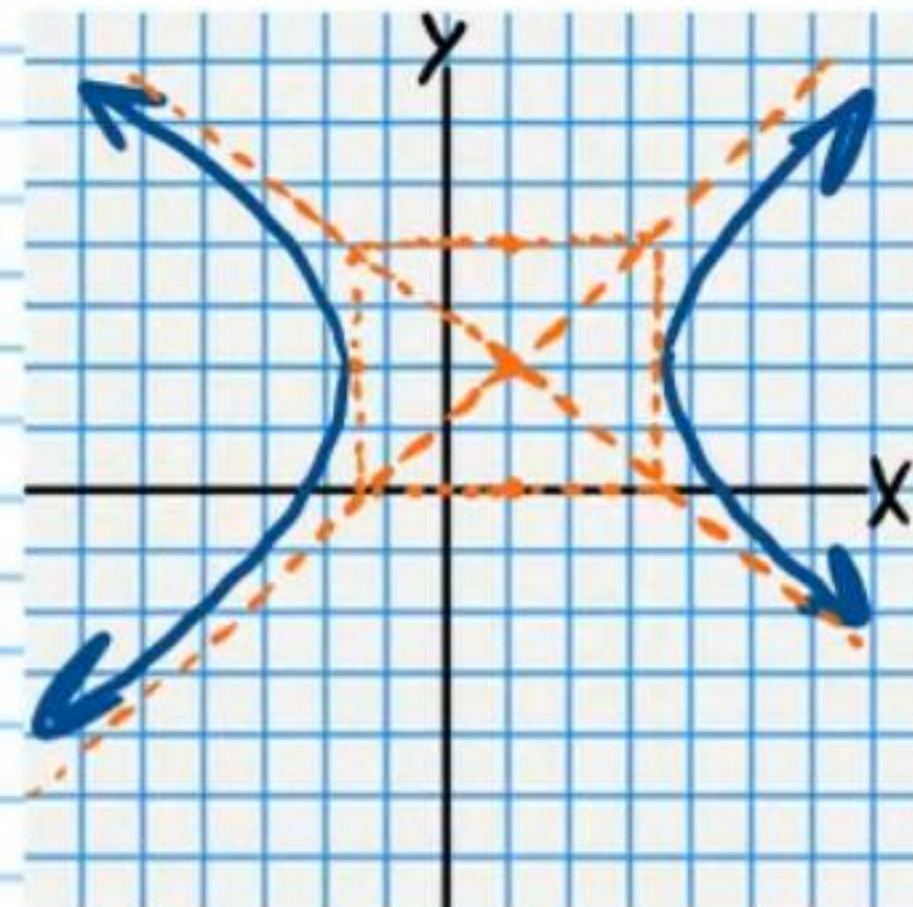
$$\frac{4(x-1)^2}{24} - \frac{6(y-2)^2}{24} = \frac{24}{24}$$

$$\frac{(x-1)^2}{6} - \frac{(y-2)^2}{4} = 1$$

$$a^2 + b^2 = c^2$$

$$(\sqrt{6})^2 + 2^2 = c^2$$

$$c = \sqrt{10}$$



Asymptotes $P(1, 2)$

$$y - y_p = m(x - x_p)$$

$$y - 2 = -\frac{2}{\sqrt{6}}(x - 1) \quad y - 2 = \frac{2}{\sqrt{6}}(x - 1)$$

$$y - 2 = -\frac{\sqrt{6}}{3}(x - 1)$$

$$y - 2 = \frac{\sqrt{6}}{3}(x - 1)$$

$$F_1(1 + \sqrt{10}, 2) \quad F_2(1 - \sqrt{10}, 2)$$

21) Solve. Write the solution graphically and in interval notation.

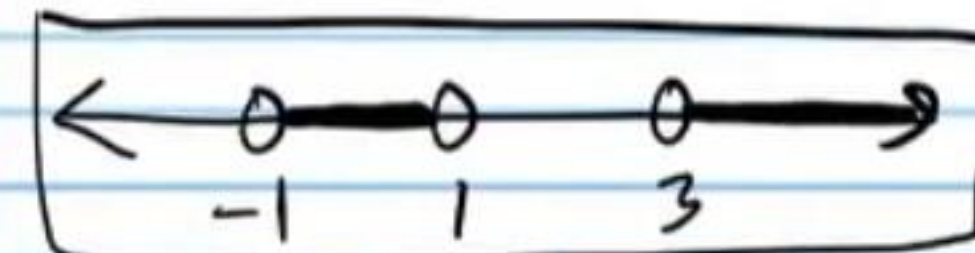
$$\frac{x^3}{x^2 - 1} - \frac{2}{x^2 - 1} > \frac{25}{x^2 - 1}$$

$$\frac{x^3 - 27}{x^2 - 1} > 0$$

$$\frac{x^3 - 3^3}{x^2 - 1} > 0$$

$$\frac{(x-3)(x^2+3x+9)}{(x-1)(x+1)} > 0$$

$$D = b^2 - 4ac = 3^2 - 4(1)(9) = -27$$



$$(-1, 1) \cup (3, \infty)$$

22) For the function, find the information requested. It is not necessary to graph the function.

$$f(x) = \frac{x^3 + 6x^2 - 4x - 24}{x+3} = x^2 + 3x - 13 + \frac{15}{x+3}$$

Asymptote Curve

$$\begin{array}{r|rrrr} -3 & 1 & 6 & -4 & -24 \\ & & -3 & -9 & 39 \\ \hline & 1 & 3 & -13 & 15 \end{array}$$

$$\boxed{y = x^2 + 3x - 13}$$

Vertical Asymptote Equation(s)

$$x = -3$$

Y-Intercept

$$f(0) = -8$$

$$\boxed{(0, -8)}$$

X-Intercept(s)

$$\frac{x^2(x+6) - 4(x+6)}{x+3}$$

$$\frac{(x+6)(x^2-4)}{x+3}$$

$$\frac{(x+6)(x-2)(x+2)}{x+3}$$

$$\boxed{(-6, 0)} \quad \boxed{(2, 0)}$$

$$\boxed{(-2, 0)}$$

23) Graph the equation below. Draw the directrix as a dotted line and write its equation. Write the coordinates of the focus.

$$-3x^2 - 18x - y - 26 = 0$$

$$-3(x^2 + 6x) = y + 26$$

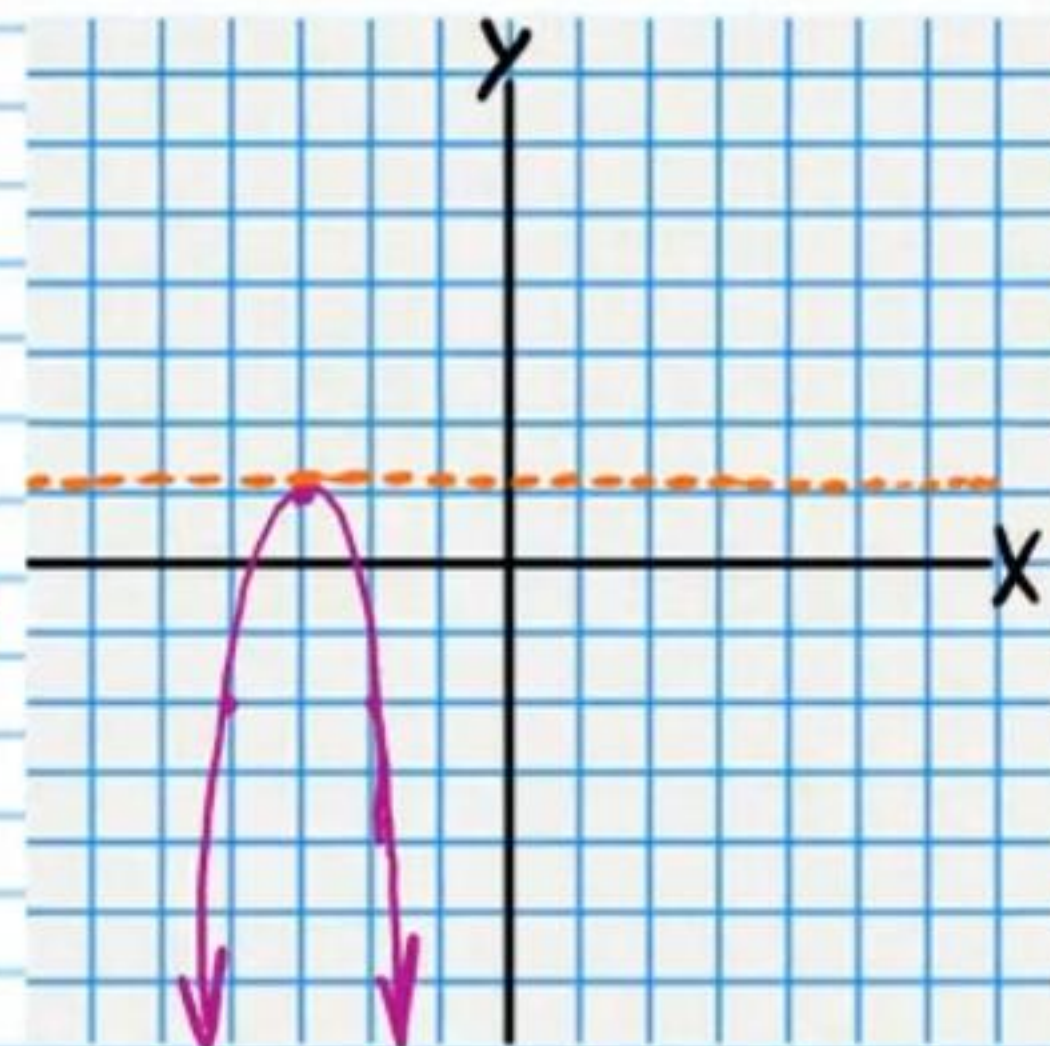
$$-3(x^2 + 6x + 9) = y + 26 - 27$$

$$-3(x+3)^2 = y - 1$$

$$p = \frac{1}{4a}$$

$$p = \frac{1}{4(-3)}$$

$$p = -\frac{1}{12}$$



$$y = 1 + \frac{1}{12}$$

$$\boxed{y = 1\frac{1}{12}}$$

$$F(-3, 1 - \frac{1}{12})$$

$$\boxed{F(-3, \frac{11}{12})}$$

24) Graph the following function. Draw asymptotes as dotted lines and write their equations.

$$f(x) = -\frac{x-3}{x^2+3x-18}$$

Hole

$$x=3$$

$$= -\frac{x-3}{(x-3)(x+6)}$$

$$g(x) = -\frac{1}{x+6}$$

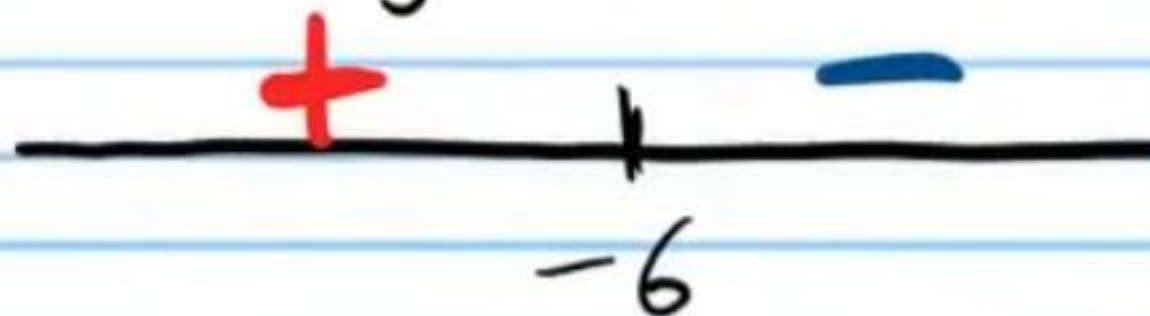
Vertical Asymptote Equation(s)

$$x = -6$$

Horizontal Asymptote Equation(s)

$$y = 0$$

Sign Chart



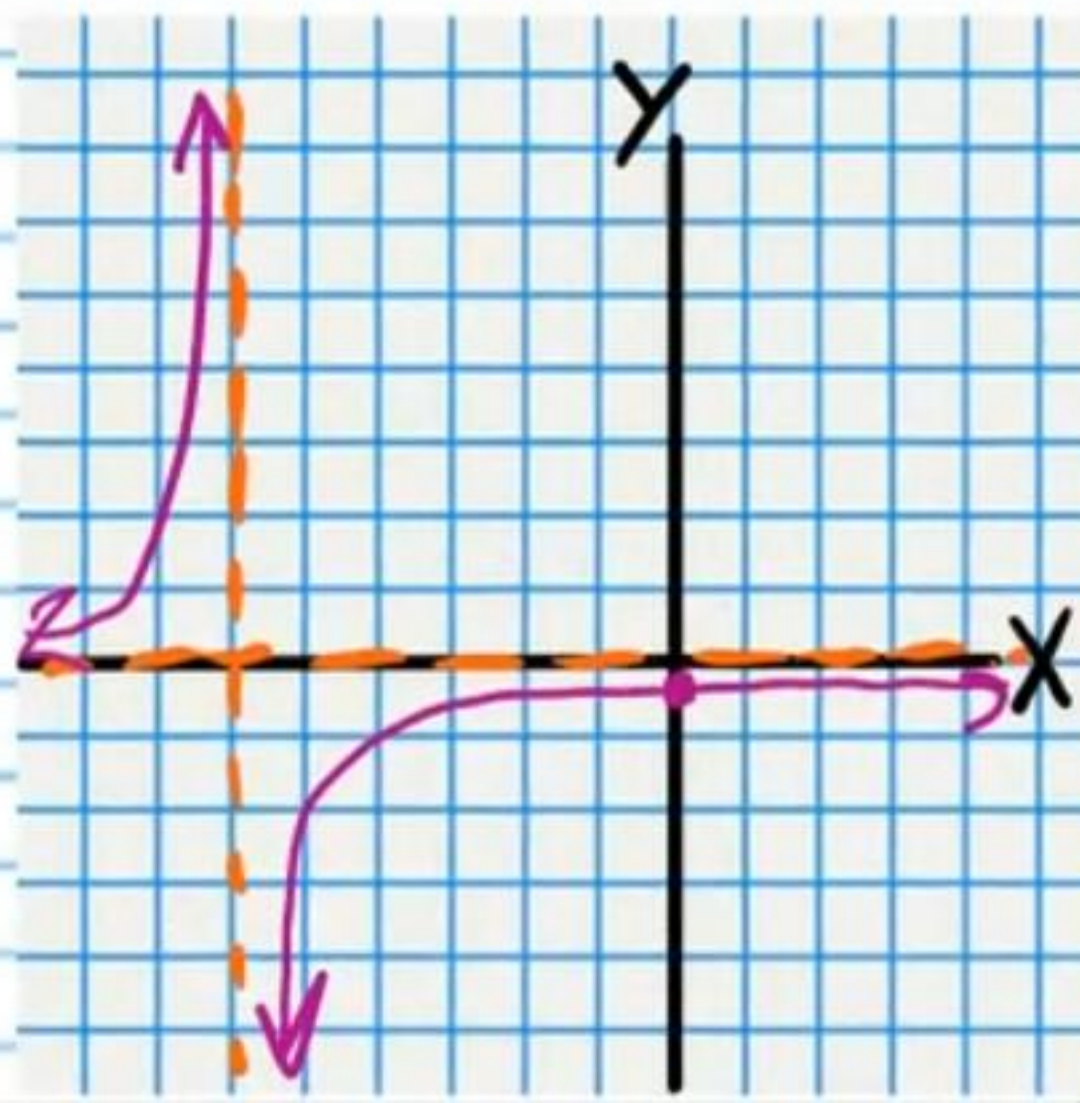
Y-Intercept

$$f(0) = -\frac{1}{6}$$

$$(0, -\frac{1}{6})$$

X-Intercept

None



25) Solve. Write the solution graphically and in interval notation.

$$x^6 + 14x + 12 \leq 7x^4 + 8x^3$$

$$x^6 - 7x^4 - 8x^3 + 14x + 12 \leq 0$$

$$p: 12 \Rightarrow 1, 2, 3, 4, 6, 12$$

$$q: 1 \Rightarrow 1$$

$$\frac{p}{q}: \pm 1, \pm 2, \pm 3, \dots$$

$$1 \Rightarrow \text{X}$$

$$-1 \Rightarrow \checkmark$$

$$\begin{array}{r|rrrrrrrr} -1 & 1 & 0 & -7 & -8 & 0 & 14 & 12 \\ & & -1 & 1 & 6 & 2 & -2 & -12 \\ \hline & 1 & -1 & -6 & -2 & 2 & 12 & 0 \end{array}$$

$$x^5 - x^4 - 6x^3 - 2x^2 + 2x + 12$$

$$-1 \Rightarrow \text{X}$$

$$2 \Rightarrow \text{X}$$

$$\begin{array}{r|rrrrrrrr} -2 & 1 & -1 & -6 & -2 & 2 & 12 \\ & & -2 & 6 & 0 & 4 & -12 \\ \hline & 1 & -3 & 0 & -2 & 6 & 0 \end{array}$$

$$x^4 - 3x^3 - 2x + 6$$

$$-2 \Rightarrow \text{X}$$

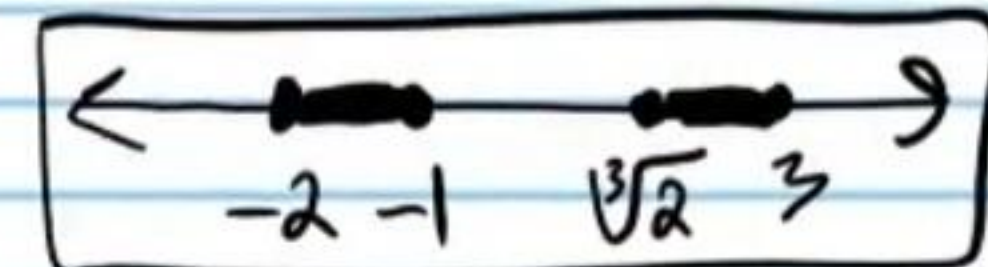
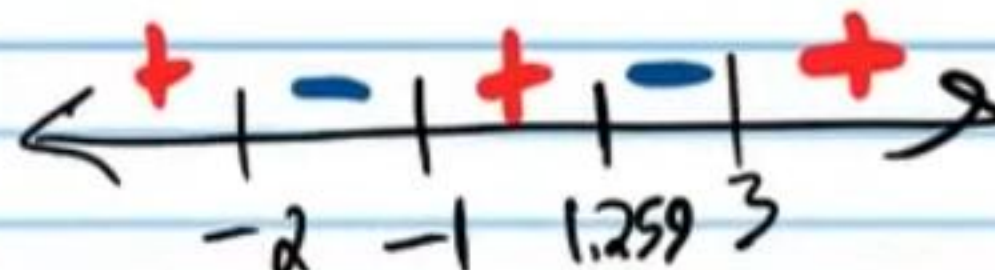
$$3 \Rightarrow \checkmark$$

$$\begin{array}{r|rrrrrrrr} 3 & 1 & -3 & 0 & -2 & 6 \\ & & 3 & 0 & 0 & -6 \\ \hline & 1 & 0 & 0 & -2 & 0 \end{array}$$

$$x^3 - 2 = 0$$

$$x = \sqrt[3]{2}$$

$$(x+1)(x+2)(x-3)(x-\sqrt[3]{2}) \leq 0$$



$$[-2, -1] \cup [\sqrt[3]{2}, 3]$$